Target validation paths — gbm-mtap-cdkn2a-idhwt-subtotal-c4bq

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

Four essential gating workups load the front-line conversation in parallel: MGMT promoter methylation by pyrosequencing (gates the adjuvant temozolomide commitment in the Stupp regimen and assigns Arm A vs Arm B at the BGB-58067 trial via NCT07485049), central MTAP IHC by the SP347-class clone (gates BGB-58067 via NCT07485049 and navlimetostat / BMS-986504 via NCT07492680), EGFR amplification by FISH plus EGFRvIII-specific RT-PCR or RNA-based fusion NGS (gates EGFRvIII-directed CAR-T families including E-SYNC via NCT06186401 and CARv3-TEAM-E via NCT05660369), and a comprehensive tumor NGS panel covering TERT, +7/-10, BRAF V600E, NTRK fusions, MMR, TMB, and MSI (gates dabrafenib + trametinib for BRAF V600E via NCT02034110, larotrectinib for NTRK via NCT02122913, and tumor-agnostic pembrolizumab for MSI-H or TMB-H via NCT02628067). Two high-priority companions sit alongside: RB1 functional IHC and p16 IHC. The Heidelberg methylation classifier is the orthogonal subtype call. All assays run on archival FFPE from the subtotal-resection block plus a peripheral blood draw for the germline panel. Turnaround is five to seven business days for the IHC bundle, seven to ten days for MGMT pyrosequencing, and two to four weeks for the comprehensive NGS.

MTAP

Essential: MTAP IHC (clone 2G4 or EPR6358) on archival FFPE, scored as complete absence of cytoplasmic staining in tumor cells with retained internal control in stromal lymphocytes and endothelium. This is the central confirmation that the MTA-cooperative PRMT5 inhibitor protocols now hard-require after MTAPESTRY-101 enrolled four of six patients on local CDKN2A-only NGS calls who turned out to have intact MTAP on central IHC and progressed on therapy. Tumor NGS copy-number at 9p21 is the wrong resolution because the signal-to-noise call rate is low and CDKN2A loss does not always extend through MTAP. Skipping the IHC risks enrolling on a non-target and forfeiting the highest-yield molecular axis in this case. The same standard carries to BGB-58067 via NCT07485049 and to navlimetostat via NCT07492680.

Medium priority: a baseline plasma SDMA draw by LC-MS/MS before starting any PRMT5 inhibitor, with a paired on-treatment draw at cycle 2 day 1 if a trial is initiated. SDMA is the most proximal pharmacodynamic readout of PRMT5 inhibition; the BMS-986504 phase 1 reported a 56% median plasma SDMA reduction from baseline to cycle 2 day 1 in MTAP-deleted responders. Most PRMT5 inhibitor trials collect this on protocol so the row often becomes a confirmation that the trial central lab is doing the draw rather than a separate send-out.

Low priority: MAT2A protein expression by IHC on archival FFPE. Research-grade antibody, useful as a translational baseline for interpreting MAT2A-inhibitor target context (IDE397, AG-270) but not required for enrollment.

CDKN2A

High priority: RB1 status combining the NGS variant call with RB protein IHC (clone G3-245). CDK4/6 inhibitors only work when RB is intact, because the kinase substrate that CDK4/6 phosphorylates is gone if RB is lost. The abemaciclib recurrent-GBM phase 2 (NCT02981940) explicitly required CDKN2A/B loss with intact RB as the eligibility pair. Tumor NGS that reports RB1 wild-type at the DNA level still misses cases where RB protein is lost post-transcriptionally, so the IHC corroboration locks the pathway integrity before the abemaciclib conversation at recurrence has anywhere to land.

High priority: p16 IHC (clone E6H4 or JC8) with the homozygous-loss pattern defined as complete absence of nuclear and cytoplasmic staining in tumor cells with retained internal control. The orthogonal protein-level call is what most CDK4/6 inhibitor protocols and tumor boards expect when the rationale rests on a single NGS copy-number signal. The IHC also catches cases where CDKN2A is functionally silenced by promoter methylation without a DNA-level deletion, which a copy-number-only panel misses.

Medium priority: CDK4 and CCND1/2/3 amplification status off the same comprehensive NGS panel. CDK4 amplification or CCND1/2/3 amplification co-occurring with CDKN2A loss creates a stronger pharmacodynamic substrate for CDK4/6 inhibition than CDKN2A loss alone; several CDK4/6 inhibitor protocols stratify or prioritize the compound genotype. Marginal cost is zero since the panel is being run anyway.

Medium priority: germline panel covering CDKN2A (familial melanoma-astrocytoma syndrome), MLH1, MSH2, MSH6, PMS2 (Lynch and CMMRD), TP53 (Li-Fraumeni), NF1, and the DNA-repair genes triggered by any tumor-side MMR loss. The homozygous-loss tumor finding is the kind of result that warrants asking whether one of the hits is germline. A high-penetrance germline call reframes surveillance and triggers cascade testing of first-degree relatives.

MGMT

Essential: MGMT promoter methylation by pyrosequencing on four to five CpG sites in exon 1, with the cutoff at approximately 8-10% average methylation and reflex to methylation-specific PCR if pyrosequencing fails. MGMT methylation is the strongest predictor of temozolomide benefit in IDH-wildtype GBM. Methylated tumors derive a 7-8 month median survival advantage on Stupp; unmethylated tumors derive limited TMZ benefit and become priority candidates for TMZ-sparing or TMZ-alternative front-line trials. Pyrosequencing was the most prognostic of the available methods in the Reifenberger and Quillien comparisons. The case has no MGMT call recorded; without it the adjuvant TMZ commitment is being made blind, the CeTeG / NOA-09 (lomustine + TMZ + RT via NCT01149109) conditional cannot open, and the patient cannot be triaged into Arm A versus Arm B at BGB-58067 via NCT07485049.

EGFR / EGFRvIII

Essential: EGFR amplification by FISH or NGS copy-number plus EGFRvIII-specific testing by RT-PCR or RNA-based NGS fusion panel, reporting amplification ratio and vIII status separately. EGFR amplification is present in roughly 40-50% of IDH-wildtype GBM (INTELLANCE screening cohort), and EGFRvIII appears in approximately 20-30% of EGFR-amplified tumors. The vIII call is the binary gate for EGFRvIII-directed CAR-T trial enrollment (E-SYNC via NCT06186401, CARv3-TEAM-E via NCT05660369, the Penn dual-CAR family via NCT07209241). Pan-EGFR amplification without vIII opens a separate set of EGFR ADC and bispecific protocols. Without resolution the patient is invisible to the most active GBM cellular-therapy trial families.

Comprehensive NGS

Essential: comprehensive tumor NGS covering TERT promoter, chromosome 7 and chromosome 10 copy-number, BRAF V600E, NTRK1/2/3 fusions, MMR genes (MLH1, MSH2, MSH6, PMS2), TMB, MSI, plus IDH1/2, ATRX, EGFR, PTEN, NF1, PDGFRA, and MET. Six high-priority biomarkers sit on one panel and each opens a different therapeutic branch: BRAF V600E gates dabrafenib + trametinib via NCT02034110 (ROAR HGG cohort, ORR 33%); NTRK fusions gate larotrectinib via NCT02122913 with confirmed CNS responses; MMR loss or MSI-H gates tumor-agnostic pembrolizumab via NCT02628067 (KEYNOTE-158, ORR 29% in TMB-H); TERT and +7/-10 confirm molecular GBM classification under cIMPACT-NOW Update 3. The same panel resolves the

EGFR amplification context and the CDKN2A homozygous-loss call, and returns a TMB number for the ICI rationale. Order RNA-based fusion testing alongside the DNA panel; CNS NTRK breakpoints are often intron-deep and DNA-only panels miss them. If MMR returns abnormal on the tumor side, escalate to a germline Lynch or CMMRD panel and trigger cascade testing.

Methylation classifier

High priority: whole-genome DNA methylation profiling with the Heidelberg / DKFZ CNS tumor classifier (Illumina EPIC array, v12.8 or later), reporting methylation class, subclass score, and MGMT promoter methylation from the same array. The classifier resolves IDH-wildtype GBM into RTK I (pediatric-type), RTK II (the most common adult subtype, enriched for 9p21 / CDKN2A homozygous loss and EGFR amplification), and mesenchymal subclasses. This case sits in RTK II genomic territory and the classifier confirms it. cIMPACT-NOW Update 9 recommends methylation profiling for high-grade glioma cases with diagnostic ambiguity; the classifier impacted the final diagnosis in 46.7% of high-confidence cases in the multicentric validation. The same array returns an orthogonal MGMT call from a different probe set than pyrosequencing, which is useful when the pyrosequencing result lands near the 8-10% cutoff.

Where to order these assays

Assay	Provider	Decision gated	Contact
MTAP IHC (clone 2G4 or EPR6358)	**NeoGenomics Laboratories (preferred) (MTAP IHC)**	**AMG 193 (MTAPESTRY 101 / NCT05094336) and other MTA-cooperative PRMT5 inhibitor and MAT2A inhibitor trial enrollment in MTAP-null GBM.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
MTAP IHC (clone 2G4 or EPR6358)	Foundation Medicine (MTAP IHC (companion to FoundationOne CDx))	AMG 193 (MTAPESTRY 101 / NCT05094336) and other MTA-cooperative PRMT5 inhibitor and MAT2A inhibitor trial enrollment in MTAP-null GBM.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
MTAP IHC (clone 2G4 or EPR6358)	Caris Life Sciences (MI Profile + MTAP IHC)	AMG 193 (MTAPESTRY 101 / NCT05094336) and other MTA-cooperative PRMT5 inhibitor and MAT2A inhibitor trial enrollment in MTAP-null GBM.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
MTAP IHC (clone 2G4 or EPR6358)	Memorial Sloan Kettering Diagnostic Molecular Pathology	AMG 193 (MTAPESTRY 101 / NCT05094336) and other MTA-cooperative PRMT5 inhibitor and MAT2A inhibitor trial enrollment in MTAP-null GBM.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
MTAP IHC (clone 2G4 or EPR6358)	Mayo Clinic Laboratories (MTAP IHC)	AMG 193 (MTAPESTRY 101 / NCT05094336) and other MTA-cooperative PRMT5 inhibitor and MAT2A inhibitor trial enrollment in MTAP-null GBM.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710

MGMT promoter methylation by pyrosequencing	**Labcorp Oncology (preferred) (MGMT Gene Methylation Assay (quantitative MSP))**	**Adjuvant temozolomide commitment (Stupp); triage into MGMT-stratified front-line trials including TMZ-sparing arms for unmethylated tumors.**	**test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363**
MGMT promoter methylation by pyrosequencing	NeoGenomics Laboratories (MGMT Promoter Methylation Analysis (bisulfite + qPCR))	Adjuvant temozolomide commitment (Stupp); triage into MGMT-stratified front-line trials including TMZ-sparing arms for unmethylated tumors.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
MGMT promoter methylation by pyrosequencing	Mayo Clinic Laboratories (MGMT Promoter Methylation, Pyrosequencing)	Adjuvant temozolomide commitment (Stupp); triage into MGMT-stratified front-line trials including TMZ-sparing arms for unmethylated tumors.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
MGMT promoter methylation by pyrosequencing	University of Michigan MLabs (MGMT Promoter Methylation)	Adjuvant temozolomide commitment (Stupp); triage into MGMT-stratified front-line trials including TMZ-sparing arms for unmethylated tumors.	test info · 1500 East Medical Center Drive, Ann Arbor, MI 48109 · 1-800-862-7284
MGMT promoter methylation by pyrosequencing	Quest Diagnostics (MGMT Promoter Methylation)	Adjuvant temozolomide commitment (Stupp); triage into MGMT-stratified front-line trials including TMZ-sparing arms for unmethylated tumors.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
EGFR amplification + EGFRvIII (FISH or NGS plus RT-PCR / RNA fusion)	**Foundation Medicine (preferred) (FoundationOne CDx + FoundationOne Heme (RNA fusion))**	**EGFRvIII-directed CAR-T and bispecific trial enrollment (E-SYNC, tandem EGFRvIII/IL-13Ra2, oncolytic-virus platforms); EGFR-amplification ADC and bispecific protocols.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
EGFR amplification + EGFRvIII	NeoGenomics Laboratories (EGFR FISH + EGFRvIII RT-PCR)	EGFRvIII-directed CAR-T and bispecific trial enrollment (E-SYNC, tandem EGFRvIII/IL-13Ra2, oncolytic-virus platforms); EGFR-amplification ADC and bispecific protocols.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
EGFR amplification + EGFRvIII	Tempus Labs (Tempus xT CDx + xR (RNA))	EGFRvIII-directed CAR-T and bispecific trial enrollment (E-SYNC, tandem EGFRvIII/IL-13Ra2, oncolytic-virus platforms); EGFR-amplification ADC and bispecific protocols.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137

EGFR amplification + EGFRvIII	Caris Life Sciences (MI Cancer Seek + WTS)	EGFRvIII-directed CAR-T and bispecific trial enrollment (E-SYNC, tandem EGFRvIII/IL-13Ra2, oncolytic-virus platforms); EGFR-amplification ADC and bispecific protocols.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
EGFR amplification + EGFRvIII	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT + MSK-Fusion)	EGFRvIII-directed CAR-T and bispecific trial enrollment (E-SYNC, tandem EGFRvIII/IL-13Ra2, oncolytic-virus platforms); EGFR-amplification ADC and bispecific protocols.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Comprehensive tumor NGS panel (TERT, +7/-10, BRAF, NTRK, MMR, TMB, MSI, IDH, ATRX, EGFR, PTEN, NF1, PDGFRA, MET)	**Foundation Medicine (preferred) (FoundationOne CDx)**	**Dabrafenib + trametinib (BRAF V600E), larotrectinib / entrectinib (NTRK fusion), tumor-agnostic pembrolizumab (TMB-H or MSI-H / dMMR), and confirmatory molecular-GBM classification.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
Comprehensive tumor NGS panel	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT (505-gene))	Dabrafenib + trametinib (BRAF V600E), larotrectinib / entrectinib (NTRK fusion), tumor-agnostic pembrolizumab (TMB-H or MSI-H / dMMR), and confirmatory molecular-GBM classification.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Comprehensive tumor NGS panel	Tempus Labs (Tempus xT CDx + xR)	Dabrafenib + trametinib (BRAF V600E), larotrectinib / entrectinib (NTRK fusion), tumor-agnostic pembrolizumab (TMB-H or MSI-H / dMMR), and confirmatory molecular-GBM classification.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
Comprehensive tumor NGS panel	Caris Life Sciences (MI Cancer Seek + WTS)	Dabrafenib + trametinib (BRAF V600E), larotrectinib / entrectinib (NTRK fusion), tumor-agnostic pembrolizumab (TMB-H or MSI-H / dMMR), and confirmatory molecular-GBM classification.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
Comprehensive tumor NGS panel	NeoGenomics Laboratories (NeoTYPE Solid Tumor Profile)	Dabrafenib + trametinib (BRAF V600E), larotrectinib / entrectinib (NTRK fusion), tumor-agnostic pembrolizumab (TMB-H or MSI-H / dMMR), and confirmatory molecular-GBM classification.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
RB1 status (NGS variant call plus RB IHC clone G3-245)	**NeoGenomics Laboratories (preferred) (RB IHC)**	**Abemaciclib (CNS-penetrant) and other CDK4/6 inhibitor eligibility in recurrent/progressive GBM with CDKN2A loss; protocols explicitly require intact RB.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**

RB1 status	Mayo Clinic Laboratories (RB IHC)	Abemaciclib (CNS-penetrant) and other CDK4/6 inhibitor eligibility in recurrent/progressive GBM with CDKN2A loss; protocols explicitly require intact RB.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
RB1 status	Labcorp / Esoterix (RB IHC)	Abemaciclib (CNS-penetrant) and other CDK4/6 inhibitor eligibility in recurrent/progressive GBM with CDKN2A loss; protocols explicitly require intact RB.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
RB1 status	Memorial Sloan Kettering Pathology Consultation Service	Abemaciclib (CNS-penetrant) and other CDK4/6 inhibitor eligibility in recurrent/progressive GBM with CDKN2A loss; protocols explicitly require intact RB.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511
p16/INK4a IHC (clone E6H4 or JC8)	**NeoGenomics Laboratories (preferred) (p16/INK4a IHC)**	**CDK4/6 inhibitor (abemaciclib favored for CNS penetration) trial enrollment; complements the CDKN2A NGS call rather than gating it.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
p16/INK4a IHC	Mayo Clinic Laboratories (p16 IHC)	CDK4/6 inhibitor (abemaciclib favored for CNS penetration) trial enrollment; complements the CDKN2A NGS call rather than gating it.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
p16/INK4a IHC	Labcorp / Esoterix (p16 IHC)	CDK4/6 inhibitor (abemaciclib favored for CNS penetration) trial enrollment; complements the CDKN2A NGS call rather than gating it.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
p16/INK4a IHC	Memorial Sloan Kettering Pathology Consultation Service	CDK4/6 inhibitor (abemaciclib favored for CNS penetration) trial enrollment; complements the CDKN2A NGS call rather than gating it.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511
Heidelberg / DKFZ CNS methylation classifier (Illumina EPIC array v12.8+)	**NeoGenomics Laboratories (preferred) (CNS Tumor Methylation Profiling)**	**Subtype refinement (RTK I vs RTK II vs mesenchymal) for trial enrollment prioritization; orthogonal MGMT methylation call from the same array.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
Heidelberg classifier	Heidelberg Epignostix GmbH (CNS Tumor Methylation Classifier)	Subtype refinement (RTK I vs RTK II vs mesenchymal) for trial enrollment prioritization; orthogonal MGMT methylation call from the same array.	test info · Im Neuenheimer Feld 280, 69120 Heidelberg, Germany
Heidelberg classifier	Johns Hopkins Molecular Pathology (Brain Tumor Methylation Profiling)	Subtype refinement (RTK I vs RTK II vs mesenchymal) for trial enrollment prioritization; orthogonal MGMT methylation call from the same array.	test info · 1800 Orleans Street, Baltimore, MD 21287 · 1-410-955-5077

Heidelberg classifier	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK Brain Tumor Methylation Classifier)	Subtype refinement (RTK I vs RTK II vs mesenchymal) for trial enrollment prioritization; orthogonal MGMT methylation call from the same array.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Heidelberg classifier	Mayo Clinic Laboratories (Brain Tumor Methylation Profiling)	Subtype refinement (RTK I vs RTK II vs mesenchymal) for trial enrollment prioritization; orthogonal MGMT methylation call from the same array.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
CDK4 / CCND1/2/3 amplification (off comprehensive NGS panel)	**Foundation Medicine (preferred) (FoundationOne CDx)**	**CDK4/6 inhibitor trial prioritization for the compound CDKN2A-loss + CDK4-amp genotype; complements CDKN2A loss rather than gating CDK4/6 inhibitor eligibility on its own.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
CDK4 / CCND1/2/3 amplification	Tempus Labs (Tempus xT CDx)	CDK4/6 inhibitor trial prioritization for the compound CDKN2A-loss + CDK4-amp genotype; complements CDKN2A loss rather than gating CDK4/6 inhibitor eligibility on its own.	test info · 600 W Chicago Avenue, Chicago, IL 60654 · 1-800-739-4137
CDK4 / CCND1/2/3 amplification	Caris Life Sciences (MI Cancer Seek)	CDK4/6 inhibitor trial prioritization for the compound CDKN2A-loss + CDK4-amp genotype; complements CDKN2A loss rather than gating CDK4/6 inhibitor eligibility on its own.	test info · 4610 South 44th Place, Suite 100, Phoenix, AZ 85040 · 1-888-979-8669
CDK4 / CCND1/2/3 amplification	Memorial Sloan Kettering Diagnostic Molecular Pathology (MSK-IMPACT)	CDK4/6 inhibitor trial prioritization for the compound CDKN2A-loss + CDK4-amp genotype; complements CDKN2A loss rather than gating CDK4/6 inhibitor eligibility on its own.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Plasma SDMA baseline (LC-MS/MS)	**Trial central laboratory (preferred) (per protocol)**	**On-treatment PRMT5-inhibitor target-engagement monitoring; informs whether to push dose or switch agents at first scan if SDMA reduction is inadequate.**	**per trial protocol**
Plasma SDMA baseline	ARUP Laboratories (SDMA, plasma)	On-treatment PRMT5-inhibitor target-engagement monitoring; informs whether to push dose or switch agents at first scan if SDMA reduction is inadequate.	test info · 500 Chipeta Way, Salt Lake City, UT 84108 · 1-800-522-2787
Plasma SDMA baseline	Mayo Clinic Laboratories (SDMA, plasma)	On-treatment PRMT5-inhibitor target-engagement monitoring; informs whether to push dose or switch agents at first scan if SDMA reduction is inadequate.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710

Germline panel (CDKN2A FAMMM, Lynch / CMMRD, Li-Fraumeni, NF1)	**Labcorp Genetics (formerly Invitae) (preferred) (Invitae Multi-Cancer Panel)**	**Cascade testing of first-degree relatives if a high-penetrance germline variant is identified; reframes surveillance recommendations for the patient.**	**test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037**
Germline panel	GeneDx (GeneDx Comprehensive Common Cancer Panel)	Cascade testing of first-degree relatives if a high-penetrance germline variant is identified; reframes surveillance recommendations for the patient.	test info · 207 Perry Parkway, Gaithersburg, MD 20877 · 1-888-729-1206
Germline panel	Ambry Genetics (CancerNext-Expanded)	Cascade testing of first-degree relatives if a high-penetrance germline variant is identified; reframes surveillance recommendations for the patient.	test info · 15 Argonaut, Aliso Viejo, CA 92656 · 1-866-262-7943
Germline panel	Myriad Genetics (MyRisk Hereditary Cancer Panel)	Cascade testing of first-degree relatives if a high-penetrance germline variant is identified; reframes surveillance recommendations for the patient.	test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423
MAT2A IHC (research-grade)	**Memorial Sloan Kettering Pathology Consultation Service (preferred)**	**Translational characterization of MAT2A-inhibitor target context (IDE397, AG-270); does not gate enrollment, which runs off MTAP-null status.**	**test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511**
MAT2A IHC	Dana-Farber / Brigham and Women's Pathology	Translational characterization of MAT2A-inhibitor target context (IDE397, AG-270); does not gate enrollment, which runs off MTAP-null status.	test info · 75 Francis Street, Boston, MA 02115 · 1-617-732-7510

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements

MTAP IHC (clone 2G4 or EPR6358) with loss defined as complete absence of cytoplasmic staining in tumor cells with retained internal control in stromal	The MTAPESTRY 101 phase 1 of AMG 193 (NCT05094336) showed that 4 of 6 patients enrolled on local CDKN2A deletion calls had intact MTAP on central IHC and progressed on therapy, which made central MTAP-null confirmation by an assay specified in the protocol a hard gate for every MTA-cooperative PRMT5 inhibitor program (AMG 193, MRTX1719, TNG462) and the MAT2A inhibitor IDE397. Tumor NGS copy-number alone is the wrong resolution for enrollment because the 9p21 region has a low signal-to-noise call rate and CDKN2A loss does not always extend through MTAP. Skipping IHC risks enrolling on a non-target and forfeiting the highest-yield molecular axis in this case.	NeoGenomics Laboratories (MTAP IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	2-3 unstained slides from archival FFPE

MGMT promoter methylation by pyrosequencing (4-5 CpG sites in exon 1, cutoff approximately 8-10% average methylation) with reflex to PCR if pyrosequencing fails	MGMT promoter methylation is the strongest predictor of temozolomide benefit in IDH-wildtype GBM; methylated tumors derive a 7-8 month median-survival advantage on Stupp, unmethylated tumors derive limited TMZ benefit and become priority candidates for TMZ-sparing or TMZ-alternative front-line trials. Pyrosequencing is the most prognostic of the available methods; the Reifenberger comparison found pyrosequencing best separated patients who benefited from temozolomide, with the 8-10% CpG-average threshold defining the cutoff. The case has no MGMT call recorded; without it the adjuvant TMZ commitment is being made blind and the patient cannot be triaged into an MGMT-stratified trial.	Labcorp Oncology (MGMT Gene Methylation Assay (quantitative MSP)) · test info · 358 South Main Street, Burlington, NC 27215 · 1-800-345-4363	1 FFPE block or 5-10 unstained slides

EGFR amplification by FISH or NGS copy-number plus EGFRvIII-specific testing by RT-PCR or RNA-based NGS fusion panel; report both amplification ratio and vIII status separately	EGFR amplification is present in roughly 40-50% of IDH-wildtype GBM (INTELLANCE screening cohort) and EGFRvIII appears in approximately 20-30% of EGFR-amplified tumors; vIII status is the binary gate for EGFRvIII-directed CAR-T (including E-SYNC, tandem EGFRvIII/IL-13Ra2 CAR-T, and oncolytic-virus-delivered bispecific platforms). Pan-EGFR amplification without vIII opens a separate set of EGFR ADC and bispecific protocols. The current report has neither call. Without resolution the patient is invisible to the most active GBM cellular-therapy trial families.	Foundation Medicine (FoundationOne CDx + FoundationOne Heme (RNA fusion)) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	1 FFPE block or 10-20 unstained slides

Comprehensive tumor NGS panel covering TERT promoter, chromosome 7 / chromosome 10 copy-number, BRAF V600E, NTRK1/2/3 fusions, MMR genes TMB, MSI, plus IDH1/2, ATRX, EGFR, PTEN, NF1, PDGFRA, MET	Six high-priority biomarkers (TERT promoter, +7/-10, BRAF V600E, NTRK fusions, MMR status, comprehensive copy-number) are unresolved and each opens a different therapeutic branch: BRAF V600E gates dabrafenib + trametinib (ROAR phase 2 HGG cohort, ORR 33%), NTRK fusions gate larotrectinib / entrectinib, MMR loss or MSI-high gates tumor-agnostic pembrolizumab, and TERT/+7/-10 confirm molecular GBM classification under cIMPACT-NOW Update 3. Running these as a single comprehensive panel is far more efficient than serial single-gene tests, and the same panel resolves the EGFR amplification and vIII context, CDKN2A homozygous-loss call, and a TMB number for the ICI rationale. Skipping forfeits four tumor-agnostic therapeutic options and weakens the molecular GBM diagnosis.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	1 FFPE block or 10-20 unstained slides; archival acceptable

RB1 status (NGS variant call plus RB protein IHC with clone G3-245) to confirm intact RB pathway downstream of CDK4/6	CDK4/6 inhibitors only work when RB is intact; RB1 loss or mutation drives primary resistance because the kinase substrate that CDK4/6 phosphorylates is gone. The abemaciclib recurrent-GBM phase 2 (NCT02981940) explicitly required CDKN2A/B loss with intact RB as the eligibility pair. Tumor NGS that reports RB1 wild-type at the DNA level still misses cases where RB protein is lost post-transcriptionally, so a quick IHC corroboration locks the pathway integrity. Without it the CDK4/6 inhibitor rationale is incomplete.	NeoGenomics Laboratories (RB IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	1 unstained slide; can share the p16/MTAP block
p16/INK4a IHC (clone E6H4 or JC8) with homozygous-loss pattern defined as complete absence of nuclear and cytoplasmic staining in tumor cells with retained internal control	p16 IHC loss correlates with CDKN2A homozygous deletion at high negative predictive value, but the orthogonal protein-level call is what most CDK4/6 inhibitor protocols and tumor boards expect when the rationale rests on a single NGS copy-number signal. The IHC also catches the cases where CDKN2A is functionally silenced by promoter methylation without a DNA-level deletion, which a copy-number-only assay misses. Skipping the IHC leaves the abemaciclib / palbociclib / ribociclib rationale resting on a single panel call and weakens the case for CDK4/6 inhibitor trial enrollment.	NeoGenomics Laboratories (p16/INK4a IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	1 unstained slide from archival FFPE

Whole-genome DNA methylation profiling with the Heidelberg / DKFZ CNS tumor classifier (Illumina EPIC array, v12.8 or later)	The Heidelberg classifier resolves IDH-wildtype GBM into RTK I (pediatric-type), RTK II (the most common adult subtype, enriched for 9p21/CDKN2A homozygous loss and EGFR amplification), and mesenchymal subclasses; this case sits in the genomic territory of RTK II and the classifier confirms it. cIMPACT-NOW Update 9 (2025) now recommends methylation profiling for high-grade glioma cases with diagnostic ambiguity. The same array returns an MGMT call from a different probe set than pyrosequencing, so it doubles as orthogonal MGMT validation.	NeoGenomics Laboratories (CNS Tumor Methylation Profiling) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	1 FFPE block or 10 unstained slides
CDK4 and CCND1/2/3 amplification status on the comprehensive NGS panel with copy-number thresholds reported	CDK4 amplification or CCND1/2/3 amplification co-occurring with CDKN2A loss creates a stronger pharmacodynamic substrate for CDK4/6 inhibition than CDKN2A loss alone, and several CDK4/6 inhibitor protocols stratify or prioritize patients with the compound CDK4 amp / CDKN2A loss genotype. The call comes off the same comprehensive NGS panel that resolves EGFR and the other rows, so the marginal cost is zero.	Foundation Medicine (FoundationOne CDx) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	shared with the comprehensive NGS block

Plasma SDMA (symmetric dimethylarginine) baseline by LC-MS/MS, with paired on-treatment draw at cycle 2 day 1 if a PRMT5 inhibitor trial is started	SDMA is the most proximal pharmacodynamic readout of PRMT5 inhibition, and the BMS-986504 phase 1 reported a 56% median plasma SDMA reduction from baseline to cycle 2 day 1 in MTAP-deleted responders, with the top dose tiers reaching 60%. A baseline draw before starting any PRMT5 inhibitor gives the team a reference for on-treatment target engagement; without it the on-treatment number is uninterpretable. Most PRMT5 inhibitor trials already collect this on protocol.	Trial central laboratory (per protocol)	5-10 mL EDTA plasma
Germline panel covering CDKN2A (familial syndrome), MLH1, MSH2, MSH6, PMS2 (Lynch / CMMRD), TP53 (Li-Fraumeni), NF1	Germline CDKN2A pathogenic variants drive the familial atypical multiple-mole melanoma syndrome (FAMMM), which carries elevated melanoma and pancreatic cancer risk and a small but recognized GBM risk; the homozygous-loss tumor finding warrants asking whether one of the hits is germline. If the tumor-side MMR / MSI panel returns dMMR or MSI-H, the same germline panel triggers the Lynch / CMMRD workup that informs first-degree-relative cascade testing.	Labcorp Genetics (formerly Invitae) (Invitae Multi-Cancer Panel) · test info · 1400 16th Street, San Francisco, CA 94103 · 1-800-436-3037	5-10 mL EDTA whole blood or saliva kit

MAT2A protein expression by IHC on archival FFPE (research-grade antibody; report nuclear and cytoplasmic intensity)	MAT2A is the upstream enzyme that supplies S-adenosylmethionine (SAM) to PRMT5; in MTAP-null tumors, MTA accumulation makes PRMT5 partially dependent on MAT2A flux, which is the synthetic-lethal rationale for MAT2A inhibitors (IDE397, AG-270). Protein-level corroboration is not required for IDE397 enrollment but a baseline MAT2A protein signal helps interpret why a given patient responds or fails.	Memorial Sloan Kettering Pathology Consultation Service · test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-5511	shared with MTAP IHC block

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.